Revision Exercise Sheet

Name: ______________________________ **Team:** __________ **Date:** __________

Instructions Work individually for 22 minutes. No device. Use notes only if your instructor permits. For Section B, write exactly one valid term, symbol, operator, expression, or value. Each item is worth 2 marks (30 marks total).

Section A — Multiple Choice Tracing (5 items, 10 marks)

1. Trace the code and choose one best answer.

```
int score = 11;
 int *first = &score;
 int *second = first;
 *second = *first + 5;
 cout << score;
```

A. 11

B. 16

C. the address of score

D. undefined behavior

2. Trace the code and choose one best answer.

```
int *data = (int*)malloc(3 * sizeof *data);
 int *tmp = (int*)realloc(data, 6 * sizeof *data);
 if (tmp != NULL) data = tmp;
```

A. If realloc fails, data still refers to the original block

B. realloc always frees the original block on failure

C. tmp must replace data before checking

D. The code zero-initializes the new slots

3. Trace the code and choose one best answer.

```
int a[7] = {4, 9, 12};
 int size = 3, capacity = 7;
 for (int i=0; i<size; ++i) cout << a[i] << " ";
```

A. 4 9 12

B. 4 9 12 0 0 0 0

C. Seven undefined values

D. Nothing

4. Trace the code and choose one best answer.

```
int a[6]={2,5,8,11}; int size=4; int index=1;
 for(int i=size; i>index; --i) a[i]=a[i-1];
 a[index]=3; ++size;
```

A. [2,3,5,8,11]

B. [2,5,3,8,11]

C. [3,2,5,8,11]

D. [2,3,8,11]

5. Trace the code and choose one best answer.

```
int a[6]={7,14,21,28}; int size=4; int index=1;
 for(int i=index; i<size-1; ++i) a[i]=a[i+1];
 --size;
```

A. Logical array [7,21,28]

B. Logical array [7,14,21]

C. Logical array [14,21,28]

D. size remains 4

Section B — Gap Completion (5 items, 10 marks)

6. Complete the pointer parameter: void addOne(int ______ value) { ++(*value); }
Answer: ______________________

7. Given Student *current;, access its id field: current_____id

Answer: ______________________

8. Validate file opening: FILE *fp=fopen(path,"r"); if(fp==_____) return false;

Answer: ______________________

9. Complete the insertion bound: for(int i=size; i _____ index; --i) a[i]=a[i-1];

Answer: ______________________

10. Complete the deletion source: for(int i=index; i<size-1; ++i) a[i]=a[i ______ 1];
Answer: ______________________

Section C — Term Matching (5 items, 10 marks)

Write the letter of the best definition beside each item. Use each option once.

#	Term	Answer
11	Dangling Pointer	______
12	Memory Leak	______
13	Invariant	______
14	Read–Validate–Commit	______
15	Whole-Record Swap	______

Option	Definition
A	A condition such as 0 <= size <= capacity that must remain true at defined operation boundaries.
B	An allocation remains unreachable because its owning address was lost without releasing it.
C	A pointer still stores an address after the referred object or allocation has ended its lifetime.
D	Parse into temporary data, verify completeness/ranges/uniqueness, then publish one valid record.
E	Move every field of a record together so field relationships remain intact during sorting.

Score: _______ / 30